

Characterization of Lysosomal Hydrolases and Transporters and Their Age-Dependent Variability: Relevance to Drug Metabolism and Transport of Small Molecule and Biologic Drugs

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Lysosomes play a key role in the accumulation, catabolism, and transport of endogenous and exogenous metabolites and proteins and are involved in drug metabolism and prodrug activation. However, the protein abundance and interindividual variability of lysosomal drug-metabolizing enzymes and transporters (DMETs) remain underexplored. In this study, we performed a global proteomics analysis of the enriched human liver lysosomal fraction to characterize and annotate lysosomal proteins and compared these results with the proteomics data of hepatocyte homogenates and the liver microsomal fraction. We annotated and quantified 66 hydrolases and 41 membrane transporters in the lysosomal fractions. These included proteins involved in prodrug activation, transport, and metabolism or functioning as drug targets. After confirming the identity of lysosomal proteins, we investigated age-dependent changes in the abundance of these proteins in human hepatocytes ($n=58$) across various age groups, ranging from neonatal (0–12 days) to adulthood (>18 years). We observed age-specific variations in the expression of key hydrolases (CTSA, CTSL, NAGLU, PLD3, and GALNS) and transporters (ATP6V1B2, ATP6V1C1, TMEM63A, and SLC39A14). Together, these findings highlight the lysosomal localization of proteins involved in drug disposition and their dynamic developmental changes, providing critical insights for refining physiologically based pharmacokinetic (PBPK) models to support precision dosing and improve therapeutic outcomes in pediatric populations.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

✓ Lysosomes are known to play an important role in intracellular transport and the activation of many prodrugs and biologics, yet their protein composition and age-dependent variability remain poorly characterized.

WHAT QUESTION DID THIS STUDY ADDRESS?

✓ This study investigated which hydrolases and transporters are present in human liver lysosomes and how their abundance changes from the neonatal period through adulthood.

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

✓ The study provides the most detailed proteomic characterization to date of human liver lysosomes, identifying 66

hydrolases and 41 transporters enriched in this compartment. It also reveals significant age-dependent changes in key lysosomal proteins, offering new insight into developmental variability relevant to drug disposition.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

✓ These findings provide foundational data needed to incorporate lysosomal metabolism and transport into pediatric and adult physiologically based pharmacokinetic (PBPK) models. This will improve precision dosing, especially for lysosomotropic small molecules and ADCs, and help anticipate age-related differences in drug exposure and response.

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Lysosomes are increasingly recognized as key players in drug metabolism and transport, beyond their classical role in cellular degradation.^{1,2} Their rich enzymatic content and acidic environment enable the breakdown of various therapeutic agents, influencing drug disposition and efficacy. For example, small molecule drugs such as doxorubicin, chloroquine, and amitriptyline (Table S1) can accumulate in lysosomes due to their physicochemical properties, particularly their combined lipophilicity and weakly basic nature.^{3,4} This sequestration can affect drug bioavailability and contribute to resistance mechanisms, especially in cancer cells.⁵ Additionally, lysosomotropic basic drugs can increase lysosomal pH and precipitate drug–drug interactions with compounds that are catabolized or accumulate within lysosomes.² Lysosomes can also be involved in the activation of prodrugs. For example, the antiviral prodrug tenofovir alafenamide (TAF) is predominantly hydrolyzed by lysosomal cathepsin A⁶ and CES1⁶ in the liver, resulting in a 6.5-fold higher intracellular concentration of the active metabolite tenofovir diphosphate compared to tenofovir disoproxil fumarate (TDF).⁷ In the context of antibody–drug conjugates (ADCs),⁸ lysosomes are central to their therapeutic activity. Many ADCs are internalized into target cells and trafficked to lysosomes, where enzymatic cleavage releases their cytotoxic payloads.⁹ For example, brentuximab vedotin delivers monomethyl auristatin E (MMAE) to CD30+ lymphoma cells, while trastuzumab emtansine releases mertansine (DM1) in HER2+ breast cancer cells.⁹ The efficiency of lysosomal processing directly impacts ADC potency and safety, making lysosomal biology a critical consideration in drug design and precision pharmacology. Also, lysosomal hydrolases are responsible for the degradation and recycling of endogenous macromolecules. Cathepsins mediate the breakdown and turnover of cytosolic and membrane proteins, whereas lipases and phospholipases are involved in lipid degradation. Glycosidases facilitate the catabolism of glycoproteins and glycolipids, collectively maintaining cellular homeostasis.¹⁰ Despite their increasing appreciation in drug disposition, lysosomes remain poorly characterized in terms of the abundance and interindividual variability of drug-metabolizing enzymes and transporters. This gap limits quantitative understanding of overall lysosomal contributions to drug disposition and hinders interpretation of *in vitro* data.

Demographic factors can profoundly influence drug metabolism and transport, thereby altering pharmacokinetics (PK) and pharmacodynamics (PD). Among these factors, age-related changes (ontogeny) in drug-metabolizing enzymes and transporters (DMETs) across developmental stages from infancy to adulthood introduce variability in the PK of drugs,¹¹ which may compromise safety and efficacy in the clinic. Pediatric populations are particularly vulnerable to variable drug metabolism leading to reduced efficacy or increased safety risks, further compounded by the scarcity of robust clinical data.¹² Current pediatric dosing practices frequently rely on allometric scaling, based on body weight or size, which often fails to adequately

capture mechanistic developmental variability.¹³ To overcome these limitations, increasing emphasis is placed on incorporating quantitative age-dependent protein abundance data of drug disposition mechanisms into physiologically based pharmacokinetic (PBPK) modeling.¹⁴ Recent advancements in mass spectrometry-based proteomics have substantially enhanced the ability to quantitate robust protein abundance data from clinical specimens to be directly integrated into PBPK models for prospective prediction of drug pharmacokinetics.¹⁵ While the ontogeny of cytochrome P450 (CYP), UDP-glucuronosyltransferase (UGT) enzymes and major drug transporters is well established,^{11,16,17} information regarding the potential influence of age on the expression levels of non-CYP enzymes is limited. Among non-CYP enzymes, hydrolases constitute a major enzyme class that metabolizes drugs with diverse chemical structures, including ester, amide and glycosidic bonds, contributing to the metabolism of ~11% of marketed drugs.¹⁸ One of the challenges in studying hydrolases involves their heterogeneous subcellular localization as some of these proteins are present in the endoplasmic reticulum (e.g., CES and AADAC),¹⁸ whereas a large number of hydrolases are localized in lysosomes. The goal of this study was to characterize lysosomal hydrolases and their age-dependent changes in abundance.

Besides consideration of lysosomal drug-metabolizing enzymes, there are numerous lysosomal transporters that are also involved in drug disposition including their efficacy. For example, efflux mediated by ABCA1 and ABCA3 is implicated in the chemotherapeutic drug resistance (sunitinib and imatinib).¹⁹ SLC46A3 is a steroid conjugate and a bile acid transporter responsible for the efflux of the catabolites of non-cleavable ADCs from lysosomes to the cytoplasm.²⁰ A bispecific antibody–drug conjugate (SLC3A2/PD-L1) leverages the lysosomal membrane transporter (SLC3A2) to enhance antitumor efficacy in the treatment of solid tumors.²¹ Despite significant drug metabolism and transport activity in lysosomes, current limited commercially available *in vitro* models to study lysosomal function are not well-characterized for their protein composition and applications.²² While hepatic *in vitro* models (microsomes, hepatocytes, and tissue slices) were systematically assessed to understand the localization and abundance of Phase I and Phase II enzymes,²³ there is a need to characterize lysosomal fractions to improve our understanding of lysosome-mediated drug metabolism and transport.

The first aim of the study was to comprehensively characterize lysosomal hydrolases and transporters by comparing the proteomics profile of human liver lysosomes (HLL) against the proteomics of human hepatocytes (HH) and human liver microsomes (HLM). This assessment also elucidated the relative expression of known drug-metabolizing enzymes and transporters across these model systems. Secondly, we investigated the age-related differences (ontogeny) in lysosomal hydrolases and transporters using human hepatocytes. Our findings offer novel insights into age-dependent variations in lysosomal proteins and provide critical data to refine

pediatric PBPK models, with the ultimate aim of enhancing safety and efficacy during pediatric drug dosing.

METHODS

Global proteomics analysis of human liver lysosomes, primary human hepatocytes, and human liver microsomes

Procurement and preparation of primary human hepatocytes, human liver microsomes, and human liver lysosomes are discussed previously²⁴ and provided in supplementary information. Global proteomic profiling of trypsin-digested samples was performed on an EASY-nLC 1,200 nanoLC system coupled to a Q Exactive HF Orbitrap mass spectrometer (Thermo Scientific, San Jose, CA). Data were acquired in positive ion mode using a data-independent acquisition (DIA) scan. Samples were first loaded onto an Acclaim PepMap[™] trap column (75 μ m $\times$ 2 cm; Thermo Scientific) for desalting, followed by peptide separation on a PepMap[™] RSLC C18 analytical column (75 μ m $\times$ 25 cm, 2 μ m; Thermo Scientific). Detailed LC and MS parameters are provided in the supplementary information.

Protein quantification using the total protein approach

LC–MS raw data were processed using DIA-NN (v1.8.1) against the Homo sapiens proteome spectral library. Default DIA-NN parameters were used, including a peptide length of 7–30 amino acids, precursor charge states of 1–4, precursor m/z range of 300–1800, and fragment ion m/z range of 200–1800, with a 1% precursor-level false discovery rate (FDR). Trypsin/P was used as the protease, with allowance for two missed cleavages and up to five variable modifications. Fixed modifications included cysteine carbamidomethylation, N-terminal methionine excision, N-terminal acetylation, and methionine oxidation. The analysis was performed in robust LC (high precision) mode, with match-between-runs (MBR) enabled and retention time–dependent cross-run normalization applied:

$$[\text{Hydrolase}]_i = \frac{\text{MS intensity}_i}{\text{Total MS intensity} \times \text{MW}_i} \quad (1)$$

Quantification of hydrolases was achieved using the total protein approach (TPA),²⁵ where protein abundance (pmol/mg total protein) was calculated from unique peptide spectral intensities and the molecular weight of each protein (Eq. 1). In this equation, MS intensity corresponds to the summed spectral intensities of all peptides belonging to hydrolase *i*; the denominator is the sum of all peptide spectral intensities in the sample; and MW_{*i*} represents the molecular weight of hydrolase *i*. Finally, hydrolase concentrations in pmol/mg total protein were converted to pmol per million hepatocytes based on the viable cell count used for proteomic analysis. The 58 human hepatocyte samples used in this study were previously analyzed to investigate the ontogeny of cytochrome P450 enzymes (CYP),¹¹ aldo–keto reductases (AKRs),²⁶ transporters, and conjugating enzymes.²⁷ No sex-specific difference was observed in the abundance of 104 lysosomal proteins upon comparison of males (*N* = 22) and females (*N* = 19) (Figure S1) human hepatocytes >1 year age.

Statistical analysis

All statistical evaluations were conducted using GraphPad Prism (version 10) and Microsoft Excel. Differences in hydrolase abundance across discrete age categories were assessed with the Kruskal–Wallis test, followed by Dunn's *post hoc* multiple comparisons.

RESULTS

Identification of proteins enriched in human liver lysosomes

Global proteomics of enriched subcellular fractions (HLM and HLL) identified over 2000 proteins; however, a significant portion of these proteins represents contaminants introduced during

the enrichment process. For example, while the Human Gene Ontology (GO) database of human liver microsomes contains only 116 proteins, we detected 2023 proteins from HLM. To identify contaminant proteins, the proteome of HLL was compared against the proteomes of HH and HLM. In each comparison, the lysosome-enriched proteins were defined as proteins that were either uniquely present in HLL or at least two-fold more abundant in HLL compared to HH or HLM. Proteins common to both lysosome-enriched sets (HLL > HH and HLL > HLM) were considered potential lysosomal-origin proteins. Comparison of HLL vs HH (Figure 1a) revealed 2004 proteins in common and 224 and 1812 proteins exclusively detected in the respective fractions. Amongst the common proteins, 870 proteins were enriched (fold change (FC) > 2) in HLL (Figure 1a). Thus, the HLL-enriched set (HLL > HH) comprised 1,094 proteins (224 unique + 870 enriched). The second comparison between HLL and HLM (Figure 1b) showed 1722 proteins in common and 506 and 301 unique proteins in the respective fractions. Amongst 1722 common proteins (Figure 1b), 415 proteins were more abundant (FC > 2) in HLL. Therefore, the HLL-enriched set (HLL > HLM) included 921 proteins (506 unique + 415 enriched). Overlaying the two lysosome-enriched sets (Figure 1c) revealed 446 proteins that were enriched in HLL when compared with proteins detected in HH or HLM, representing high confidence in the lysosomal protein candidates.

Proteomic landscape of human liver lysosomes

To further verify the protein annotations, 446 lysosome-enriched proteins were queried against the Gene Ontology (GO) database for human lysosomal proteins, resulting in a match for 159 proteins (Figure 2a). Key lysosomal marker proteins CTSD, CTSD (Figure 2b), LAMP1, LAMP2, and LIMP2 were consistently identified across 13 HLL samples from 13 individual donors (Figure S2), exhibiting a coefficient of variation (CV) for relative abundance below 31% (Figure 2c), suggesting that the purification and data filtration workflow effectively retained proteins of lysosomal origin. The String analysis of molecular function annotated 66 hydrolases (GO:0016787) and 41 transporters (GO:0005215). Amongst these 66 luminal hydrolases, the major functional classes (Figure 2d) include 21 peptidases, 19 glycosidases, 19 esterases, and 7 lipases. Forty-one transporter proteins (Figure 2e) were linked to various transporter classes (e.g., SLC and ABC families), including 23 cation transporters, 8 ATPase-coupled transporters, 7 lipid transporters, and 7 anion transporters.

Qualitative and quantitative comparison of the curated lysosomal proteins between human liver lysosomes and human hepatocytes

The curated 159 HLL proteins were compared with the HH proteome to evaluate the presence of lysosomal proteins in complex HH homogenates. This assessment identified a total of 104 lysosomal proteins (Figure 3a) in HH homogenates. The remaining 55 proteins were likely below the limit of quantification in the HH homogenate. Table S4 compares the absolute abundance of 104 lysosomal proteins between HH, HLL, and HLM. The top 10 abundant lysosomal proteins in human

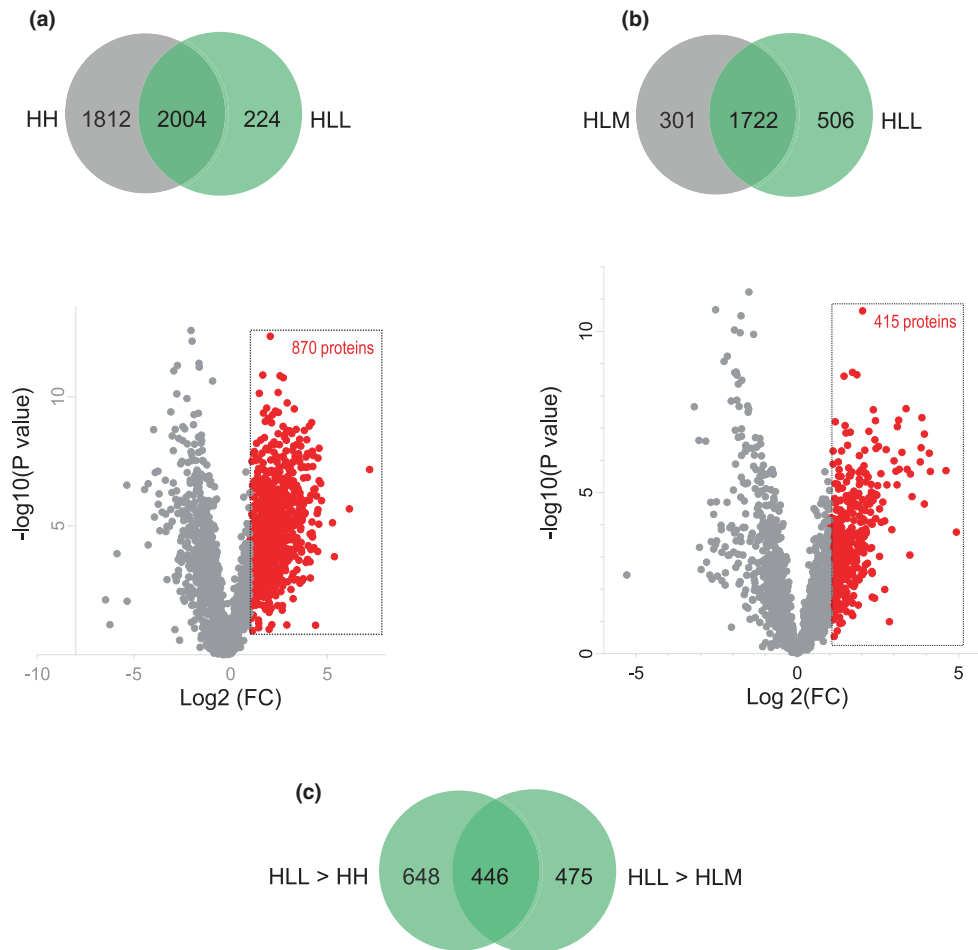


Figure 1 Identification of proteins enriched in human liver lysosomes: (a) Comparison of HLL versus HH: A Venn diagram illustrates proteins uniquely identified in HLL and HH, as well as those shared between the two. The accompanying volcano plot displays the shared proteins, highlighting those enriched more than twofold in the HLL fraction in red. (b) Comparison of HLL versus HLM: A Venn diagram shows proteins exclusively detected in each fraction and those shared between HLL and HLM. The corresponding volcano plot highlights proteins enriched more than twofold in the lysosomal (HLL) fraction in red. (c) Comparative analysis of HLL-enriched fractions illustrated by a Venn diagram of common and uniquely enriched proteins in HLL compared to either HH (HLL>HH) or HLM (HLL>HLM). Each HLL-enriched fraction includes proteins that are either exclusively detected in HLL or show more than 2-fold higher abundance in HLL relative to HH and HLM.

hepatocytes (Figure 3b) accounted for ~66% of the total lysosomal protein abundance, where the two most abundant proteins, CTSD and CTSB, represented ~41% of the total lysosomal protein abundance. A strong correlation was found in the abundance of 104 lysosomal proteins between HLL and HH (Figure 3c,d, Pearson correlation analysis, $r = 0.87$). Collectively, these findings suggest that HH can be used to investigate the ontogeny of 104 lysosomal proteins.

Age-dependent changes in lysosomal proteins

To assess age-dependent alterations in the lysosomal proteins, we compared the label-free quantification (LFQ) values normalized to the total protein abundance from human hepatocytes across six age groups: 0 to 12 days ($n = 7$), 26 days to 1 year ($n = 10$), 1 to 6 years ($n = 22$), 6 to 12 years ($n = 6$), 12 to 18 years ($n = 5$), and > 18 years ($n = 8$). The sum of 104 lysosomal proteins (Figure 4a) as a surrogate for the total lysosomal fraction proteins was higher in age groups <1 year compared to age groups >1 year.

Categorical statistical analysis of individual lysosomal proteins revealed numerous proteins exhibiting age-dependent changes in their abundance. Lysosomal proteins were grouped based on their molecular function (Figure 4b–e), which highlighted distinct shifts in the mean protein abundance across different developmental stages.

The levels of several peptidases (CTSA, CTSC, CTSL, CTSS, DPP4, LGMN, and PRCP) decreased with age, except for DPP2, which showed an increase in the protein abundance with age (Figure 4b). Two distinct ontogeny patterns were observed for glycosidases. The levels of GLB1, NAGA, and HEXA peaked in two age groups: 0–12 days or 26 days to 1 year and > 18 years. In contrast, the levels of GM2A and NAGLU gradually declined with age. Six lysosome-enriched esterases exhibited age-related changes in their protein abundance. Among them, the abundance of the lipases PLBD2 and PLD3 declined with increasing age (Figure 4d), while the levels of GALNS and SIAE increased with age (Figure 4d). Additionally, the abundances of eight

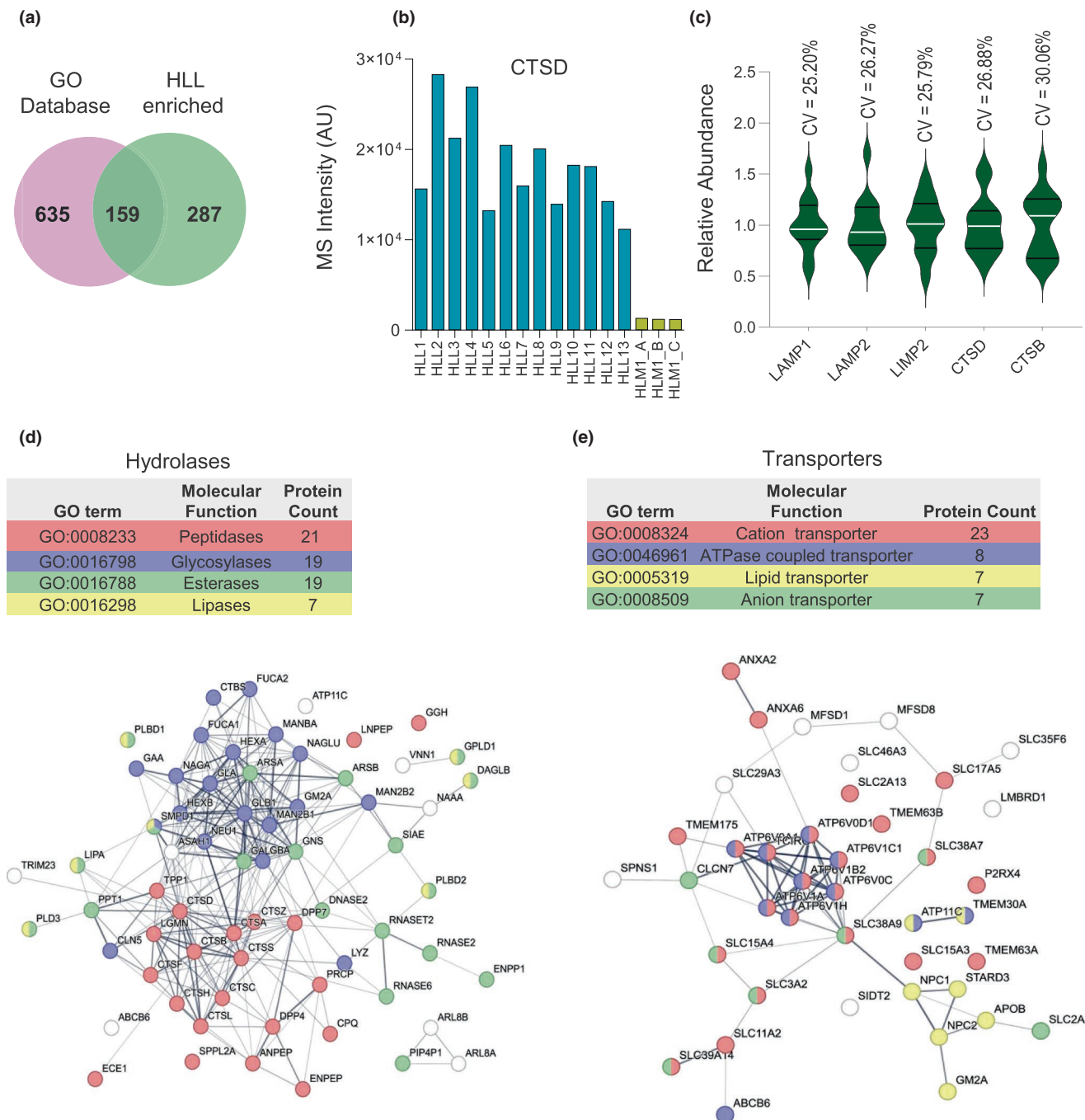


Figure 2 Annotation of molecular function of lysosomal proteins. (a) Overlap between curated lysosomal proteins and the Gene Ontology (GO) database of human lysosomal proteins. (b) Comparison of the abundance of CTSD across 13 individual HLL donor samples and a pooled HLM sample, represented by three technical replicates (HLM1_A, HLM1_B, and HLM1_C). (c) Reproducibility of key lysosomal marker proteins (LAMP1, LAMP2, LIMP2, CTSD, and CTSB) across 13 individual HLL samples. (d) STRING network analysis illustrating the interactions among luminal hydrolases categorized by molecular function. (e) STRING network analysis of lysosomal membrane transporters, classified by molecular function.

transporter proteins changed with age. The protein abundance of ATP6V1B2 and ATP6V1C1 (Figure 4e), structural components of the V-ATPase complex, decreased with age. On the other hand, the levels of three ion transporters, SLC39A14, TMEM63A, and TMEM175, increased with age.

Comparison of CYPs, non-CYPs, and Transporters in human liver lysosomes, human hepatocytes and human liver microsomes

From the 648 drug-metabolizing enzymes and transporters (DMET) proteins listed in the Human Gene Ontology (GO)

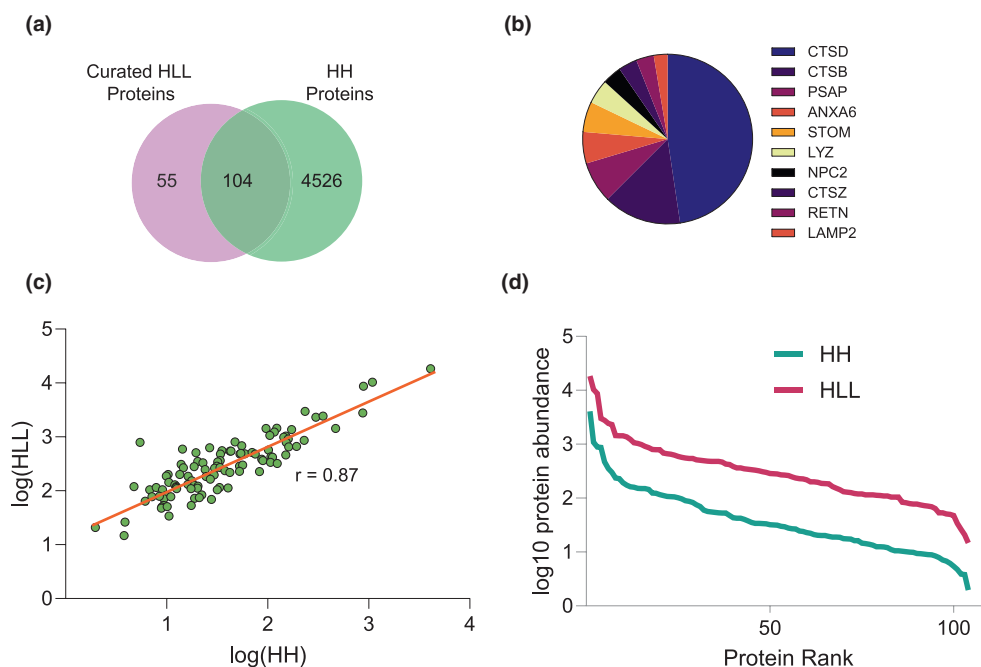


Figure 3 Qualitative and quantitative comparison of HLL and HH proteome. (a) Overlay of curated and annotated lysosomal proteins of HLL with the global proteome of HH. (b) Top 10 most abundant lysosomal proteins in HH. (c) Pearson correlation of protein intensity between HLL and HH. (d) Comparative ranking of protein abundance between HLL and HH.

database, 60 were consistently identified across the HLL, HLM, and HH proteomes (Table S5). Individual protein abundances were normalized with total protein content to enable the cross-model comparison (HLL vs HH and HLL vs HLM). The comparison of the abundance between HLL and HH (Figure 5a) revealed enrichment ($FC > 2$, 2- to 18-fold) of 43 and 6 proteins, respectively. All the CYP isoforms (e.g., CYP3A4, CYP2D6, CYP2C9, CYP2C8, and CYP2C19) and UDP-glucuronosyltransferases (e.g., UGT2B4 and UGT1A1) exhibited higher abundance in HLL relative to HH. In contrast, cytosolic Phase II enzymes SULT1A1 and SULT2A1 were abundant in HH compared to HLL. Among the five glutathione transferases, GSTA1 and GSTP1 were enriched in HLL, while GSTO1 and GSTM1 were abundant in HH. Membrane-bound transporters belonging to the SLC and ABC superfamilies were predominantly enriched in HLL compared to HH.

The comparison of DMET protein abundance between two enriched subcellular fractions, HLL and HLM (Figure 5b), showed greater abundance ($FC > 2$, 2–5-fold) of 13 and 7 proteins in the respective fractions. As anticipated, most ER-localized CYP isoforms were enriched in HLM compared to HLL. Six UGT isoforms also showed higher abundance in HLM. In contrast, sulfotransferases (SULT1A1, SULT2A1) and glutathione S-transferases (GSTA1, GSTM1, GSTM3, GSTP1, GSTO1) were abundant in HLL compared to HLM. Furthermore, many SLC and ABC transporters were enriched more abundantly in HLL than in HLM, except for SLC22A18, which is abundant in HLM.

DISCUSSION

Quantitative proteomics of DMET proteins provides an opportunity to improve scaling of *in vitro* data and to incorporate

population-based interindividual differences, thus enabling refinement of PBPK models for predicting drug pharmacokinetics. The latter is particularly valuable to improve the accuracy of pharmacokinetic predictions for populations underrepresented in clinical trials, such as children. Previous ontogeny studies primarily focused on analysis of liver microsomes and cytosols for Phase I and Phase II enzymes^{11,28} including some hydrolases²⁹; however, this leaves substantial knowledge gaps for the proteins present in other critical subcellular fractions and their contribution to metabolism and intracellular transport. In particular, new drug modalities are metabolized by diverse and sometimes atypical metabolic pathways in different subcellular fractions. For instance, lysosomes play a critical role in the activation and clearance of prodrugs and biologics such as ADCs. However, the lysosomal fraction is typically not characterized for the abundance of DMET proteins. Similarly, the ontogeny of lysosomal proteins remains underexplored. The objective of this study was to comprehensively characterize the human liver lysosomes proteome with a focus on DMETs and age-dependent changes in protein abundance across a broad developmental range.

In vitro subcellular models (HLM and HLL) provide mechanistic insights into intracellular drug processing, which is masked in whole-cell or tissue models. However, challenges remain in the enrichment and characterization of the true subcellular proteome. The relatively small mass of lysosomes² (<1–5% of total cell volume) leads to cross-contamination with other subcellular compartments and their fragments produced during the enrichment process. For example, we detected several ER-localized proteins (e.g., CYPs and UGTs) in the HLL samples. Although lysosomes represent the terminal compartment of the normal cellular degradation pathway, the levels of subcellular organelle marker proteins

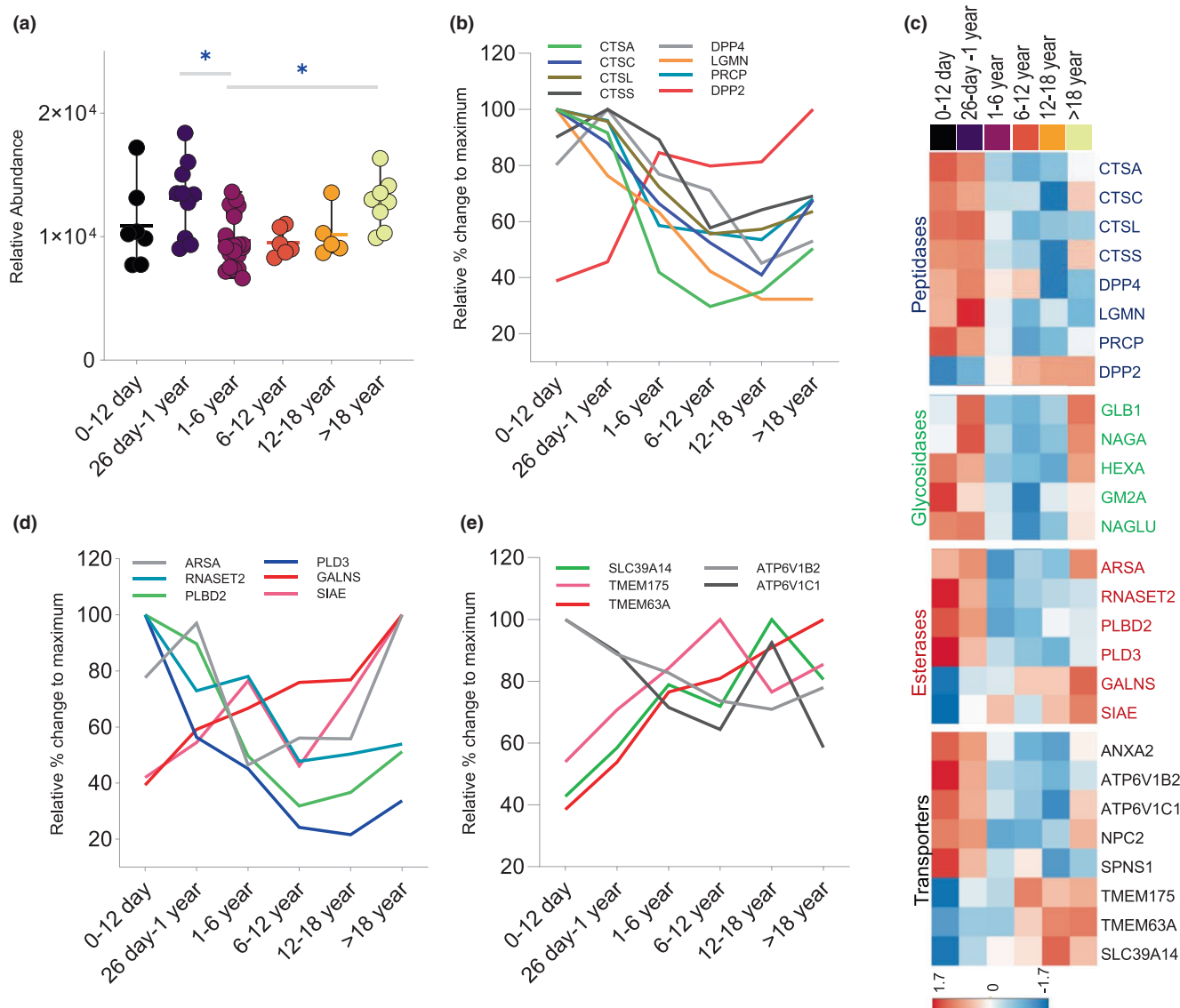


Figure 4 Ontogeny of lysosomal proteins in 58 human hepatocyte samples of age groups 0–12 days ($n=7$), 26 days–1 year ($n=10$), 1–6 years ($n=22$), 6–12 years ($n=6$), 12–18 years ($n=5$), and >18 years ($n=8$). **(a)** Ontogeny of total lysosomal protein across different age groups. **(b)** Age-dependent expression patterns of peptidases. **(c)** Heatmaps showing the mean abundance across age groups. Proteins are grouped by molecular functions, that is, peptidases, glycosidases, esterases, and transporters. **(d)** Age-dependent expression patterns of esterases. **(e)** Age-dependent expression patterns of transporters.

in HLL clearly indicate technical contamination originated during the enrichment process by HLM and other vesicular organelles. Similarly, previous studies reported co-sedimentation of ER-derived fragments in the lysosomal fraction.^{30,31} This underscores the need for careful interpretation of proteomics data of enriched subcellular fractions and implementation of rigorous filtering strategies. In the present study, to identify if proteins detected in HLL indeed originate from lysosomes, we performed a comprehensive analysis to eliminate potential contamination by comparing the proteome of HLL against the proteome of HH and HLM. This analysis led to the identification of proteins either uniquely present or significantly enriched in HLL, providing a basis for excluding proteins of non-lysosomal origin. For example, only 446 proteins passed our rigorous criteria to be considered as unique to

or enriched in lysosomes out of a total of 2,228 proteins detected in HLL. Among these, 159 proteins matched with the Human Gene Ontology (GO) database of human lysosomes, including key lysosomal marker proteins (LAMP1, LAMP2, CTSD, etc.), supporting our analysis. The remaining 287 proteins may represent liver-enriched proteins that are not fully characterized in the GO database. Nevertheless, molecular function analysis annotated 66 hydrolases and 41 transporters, which also represent lysosomal proteins involved in drug targeting, prodrug activation, and catabolism as described below. Cathepsins were amongst the most abundant class of lysosomal proteins, consistent with literature, which has been exploited in targeted drug delivery systems to facilitate the selective intracellular release of cytotoxic payloads.⁹ Valine–citru-line (Val–Cit) linker used in FDA-approved ADC brentuximab

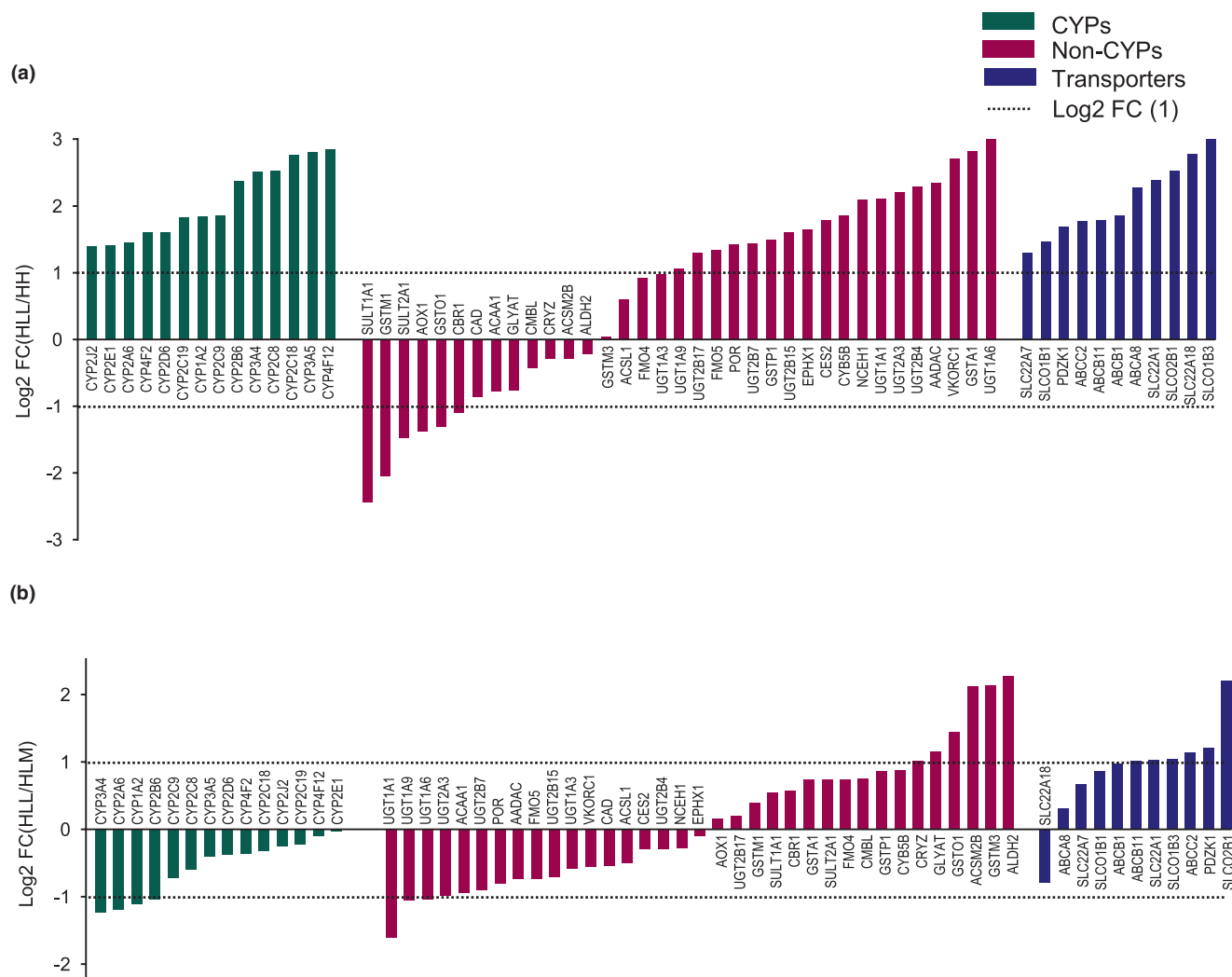


Figure 5 Comparison of the relative expression of drug-metabolizing enzymes and transporter proteins quantified from global proteomics of HLL, HLM, and HH. Data are shown in log2 fold changes, where positive values indicate higher expression in HLL and negative values indicate lower expression in HLL. (a) log2 fold change (FC) of HLL versus HH. (b) log2 FC of HLL versus HLM.

vedotin is catabolized by CTSB to release potent cytotoxic payload monomethyl auristatin E (MMAE).³² SLC46A3 (lysosomal proton-coupled steroid conjugate and bile acid transporter) mediates the efflux of catabolites of non-cleavable ADCs from the lysosome to the cytoplasm.³³ Compromised SLC46A3 function is implicated in drug resistance and cytotoxicity.³⁴ An *in vitro* study explored SLC46A3 as a potential patient selection biomarker for maytansine-based ADCs such as trastuzumab emtansine.³⁵ Enrichment of a large number of other solute carrier transporters (SLC) and ATP-binding cassette (ABC) transporters in HLL also revealed a critical, yet likely underappreciated role of lysosomes in intracellular transport and sequestration of xenobiotics. These findings also indicate the role of uptake and efflux transporters that may act in concert with hydrolases. Hydrolysis often leads to a significant change in the lipophilicity of a compound. For example, substrates of β -glucuronidase (GUS), an enzyme exclusively localized in lysosomes, are typically hydrophilic and therefore require uptake transporters for entry into the lysosomes.^{36,37} Similarly, metabolites formed by esterases tend to have reduced lipophilicity,

resulting in slower passive diffusion. Thus, efflux transporters are likely needed to aid their release from lysosomes.³⁸

While the enriched subcellular fractions enable the measurement of low-abundance proteins, their application is challenging for determining their interindividual variability due to labor-intensive preparation and potential differences in level of enrichment between preparations. To investigate the ontogeny of lysosomal proteins, we compared the annotated 159 lysosomal proteins (from HLL) with the HH proteome, revealing the presence of 104 lysosomal proteins within the complex hepatocyte homogenates. Our study showed the total lysosomal protein abundance is significantly higher in <1 year age groups (infants and neonates) compared to >1 year age groups (children, adolescents, and adults), suggesting increased lysosomal volume or activity during early stages of life, consistent with the elevated metabolic demand of rapidly dividing cells and developing organs.^{39,40} The protein abundance of several peptidases, such as CTSA, CTSC, CTSL, CTSS, DPP4, LGMN, and PRCP, decreased with age, whereas DPP2 abundance increased with age. Peptidases are leveraged in prodrug design to achieve targeted and intracellular release, thus minimizing unwanted systemic

exposure.⁴¹ For instance, CTSA catalyzes the initial hydrolytic step required to activate antiviral prodrugs (tenofovir alafenamide, sofosbuvir, and remdesivir) in the liver⁴² and lungs.⁴³ Similarly, DPP-4 is a pharmacological target, which can be inhibited by drugs referred to as gliptins (e.g., sitagliptin, linagliptin, and saxagliptin) for the treatment of type 2 diabetes mellitus.⁴⁴ The age-dependent expression of these peptidases may affect the efficacy and safety of their substrates across different age groups. We observed age-related changes in the abundance of esterases, a subclass of hydrolases involved in the metabolism of drugs such as aspirin, clopidogrel, lovastatin, and local anesthetics.⁴⁵ Esterases involved in lipid metabolism (PLBD2 and PLD3, and ARSA) are decreased with age. In contrast, the abundance of GALNS (N-acetylgalactosamine-6-sulfatase) increased with age, an enzyme responsible for the breakdown and recycling of glycosaminoglycans. Deficiency in GALNS level due to a genetic mutation implicated in a rare lysosomal storage disorder mucopolysaccharidosis IV A.⁴⁶ Similar to CYPs and UGTs,^{11,28} ontogeny of lysosomal proteins is largely non-monotonic in nature, highlighting the complex role of lysosomes in developmental regulation. ATP6V1B2 and ATP6V1C1 (subunits of the V-ATPase pump) showed an age-dependent decline in their abundance. This could potentially perturb an acidic microenvironment responsible for hydrolase activity and accumulation of basic lipophilic drugs. ATP6V1B1 mutations are linked to distal renal tubular acidosis, attributable to disruption of renal acid excretion.⁴⁷ In contrast, the protein abundance of three ion transporters, TMEM63A, TMEM175, and SLC39A14, increased with age. TMEM175 is a selective K⁺ ion channel involved in the regulation of the lysosomal acidic environment.⁴⁸ SLC39A14 primarily facilitates the transport of manganese, zinc, and iron across cellular membranes.⁴⁹

While global proteomic analyses of *in vitro* model systems provide quantitative insights and support hypothesis generation, independent mechanistic studies are required to more rigorously establish the precise localization of enzymes and transporters. Further investigations are needed to understand the functional activities and regulation mechanisms. One of the limitations in the field is the lack of physiologically relevant lysosomal models and the limited availability of pediatric liver tissue samples. Unlike CYP and UGT enzymes, robust *in vitro* models, along with well-characterized substrates and inhibitors, are needed for lysosomal hydrolases. Furthermore, potential disease comorbidities in donor hepatocytes may confound the interpretation of lysosomal protein abundance. Nevertheless, this is the first study to provide a comprehensive characterization map of the human liver lysosomal proteome and its age-dependent changes in protein abundance, revealing how hydrolases and transporters evolve from infancy to adulthood and shape intracellular drug metabolism. We elucidated age-dependent changes in key lysosomal enzymes (e.g., peptidases, hydrolases, glycosidases, and esterases) and transporters, underscoring the dynamic nature of lysosomal function across development. Integrating age-dependent lysosomal protein data into PBPK models will enable more accurate prediction of drug exposure in pediatric populations, particularly for lysosomotropic drugs and antibody-drug conjugates. With the rapid expansion of innovative and targeted therapies that increasingly depend on non-CYP metabolic pathways, this work is particularly relevant for establishing safe and effective dosing strategies in pediatric populations.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

The authors declare the following competing financial interest(s): Bhagwat Prasad (Corresponding Author) is co-founder of Precision Quantomics Inc. and recipient of research funding from Bristol-Myers Squibb, Genentech, Gilead Sciences, Merck, Novartis, Takeda, AbbVie, Boehringer Ingelheim, and Generation Bio. All other authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

D.G., D.K.S., S.S., G.Y., D.A., S.H., R.S.J., C.K., R.K., P.K., B.M., B.Mu., B.S., D.M.S., M.E.T., T.W., and B.P. wrote the manuscript and designed the research. D.G., D.K.S., S.S., G.Y., C.B., M.C., S.H. and C.S. performed the research. D.G. and B.P. analyzed the data.

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[Correction added on 06 June 2026, after first online publication: The copyright line was changed.]

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